CS683 Programming Assignment II

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All simulations use ChampSim with 25M warmup and 25M simulated instructions on the three provided traces (Trace1–Trace3). Speedup is always relative to the baseline with no prefetching and LRU at the L2C. Aggregate rows use the geometric mean for IPC and speedup, and the arithmetic mean for MPKI, accuracy and coverage.

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1 Task 3: Prefetch-Aware Dynamic I-RRIP with SPP

1.1 Design Rationale: Handling Prefetch Packets

The Task 2B policy is kept unchanged; prefetch awareness is added on top of it using the packet type (`type == PREFETCH`).

Why prefetched lines need special handling. Most SPP prefetches are filled into the L1D, so the demand access that follows is served by the L1D and the copy in the L2 is often never touched again. Our runs confirm this: of the prefetched lines filled into the L2, only 2.8–3.8% received a demand hit there before eviction (ChampSim L2C prefetch counters, e.g. 1,649 useful vs. 44,260 useless on Trace1). If these lines are treated like demand data, they push out the instruction lines that I-RRIP is trying to protect.

- **Tracking.** One extra bit per L2 line marks "brought in by a prefetch and not yet used by a demand access".
- **Insertion.** Prefetch fills are inserted according to an L2-side usefulness monitor:
  - A prefetched line is *useful* if a demand access hits it before eviction, and *useless* if it is evicted untouched.
  - Every 512 resolved prefetches, if usefulness $\geq 40\%$, prefetches are inserted at RRPV 2 (like normal data); otherwise at RRPV 3, the eviction position, so a wrong prefetch occupies a way only briefly.
- **Promotion.**
  - A `PREFETCH` request that hits a resident line does *not* promote it: a prefetch says nothing about future demand reuse.
  - The first demand hit on a prefetched line clears its bit, counts it as useful, and promotes it only to RRPV 1, since its next uses will mostly be served by the L1D. Later demand hits follow normal Dynamic I-RRIP promotion.
- **Eviction.** Among lines at max RRPV, an unused prefetched line is evicted first, then data, then instruction lines (with the 2B instruction-cap rule). Empty ways are always filled first.
- **Set-dueling monitor.** Prefetch traffic is excluded from the instruction-share formula. Otherwise heavy prefetching inflates the data side and wrongly pushes the cache into plain RRIP mode.
- **Leader sets stay pure RRIP**, so the set duel remains a fair comparison.
- **Cost.** One bit per L2 line plus a few small counters (useful/useless counts, epoch counter, current insertion RRPV).

1.2 Results

All configurations use 25M warmup + 25M simulated instructions. Baseline = no prefetching, LRU at L2C. Dynamic I-RRIP uses threshold 0.25.

Table 1: Task 3: IPC and speedup over baseline

	IPC			Speedup	
Trace	Baseline	Dyn I-RRIP	SPP + PF-aware	Dyn I-RRIP	SPP + PF-aware
Trace1	0.2323	0.2320	**0.2338**	0.9988	**1.0064**
Trace2	0.2514	0.2577	**0.2582**	1.0250	**1.0269**
Trace3	0.3819	0.3818	**0.3848**	0.9997	**1.0075**
Geomean	0.2815	0.2837	**0.2853**	1.0078	**1.0136**

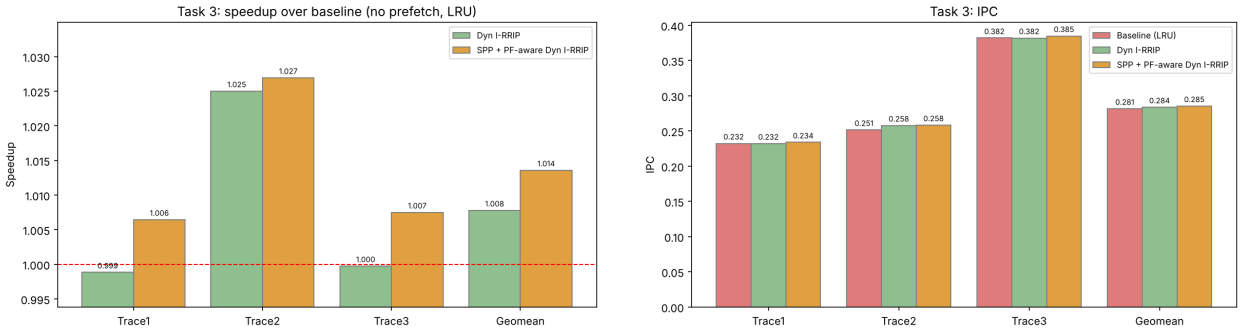
Table 2: Task 3: L1D prefetch accuracy and coverage (SPP + PF-aware Dyn I-RRIP)

Trace	Accuracy, ChampSim (%)	Accuracy, SPP filter (%)	L1D coverage (%)
Trace1	50.13	69.11	8.82
Trace2	32.12	58.91	1.25
Trace3	27.19	59.49	8.26
Mean	36.48	62.50	6.11

ChampSim accuracy = useful load prefetches / prefetches issued to the L2 (only prefetches that missed in the L1D). SPP filter accuracy = useful / all prefetches issued by SPP (its internal throttling metric). Coverage = (baseline L1D load misses − L1D load misses with SPP) / baseline L1D load misses.

Table 3: Task 3: load MPKI (lower is better). B = baseline, D = Dyn I-RRIP, P = SPP + PF-aware Dyn I-RRIP

	L1D Load			L2C Load			L2 Instruction Load			L2 Data Load		
Trace	B	D	P	B	D	P	B	D	P	B	D	P
Trace1	10.68	10.68	**9.74**	17.68	17.74	**16.89**	10.35	9.92	9.92	7.33	7.82	**6.97**
Trace2	13.54	13.54	**13.37**	27.35	21.54	**21.44**	18.78	12.44	12.55	8.56	9.10	8.90
Trace3	3.40	3.40	**3.12**	3.96	3.90	**3.40**	1.13	1.01	1.01	2.83	2.89	**2.39**
Mean	9.21	9.21	**8.74**	16.33	14.39	**13.91**	10.09	7.79	7.82	6.24	6.60	**6.09**



(a) Speedup over baseline

(b) IPC

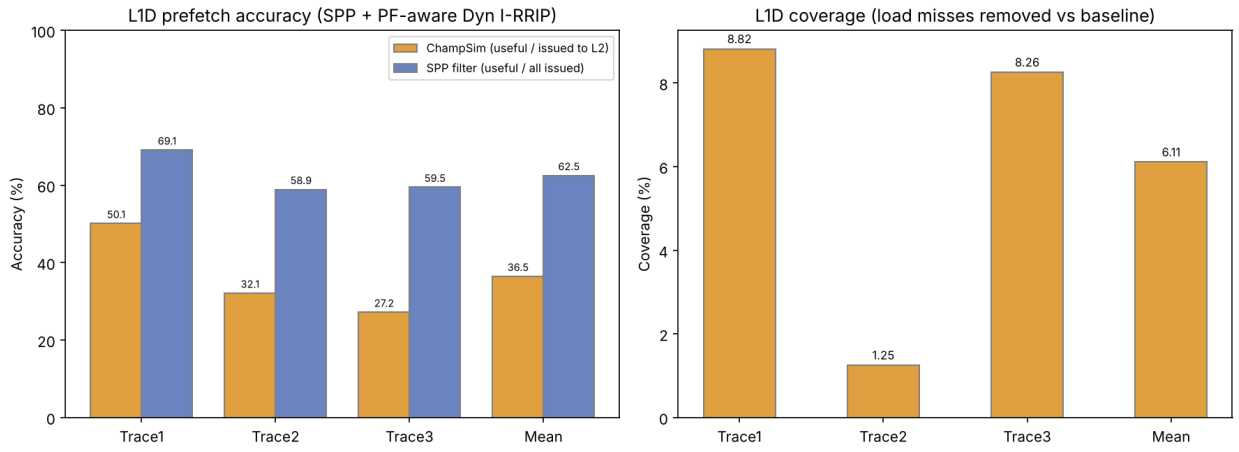


Figure 1: Task 3: performance, prefetch accuracy and coverage

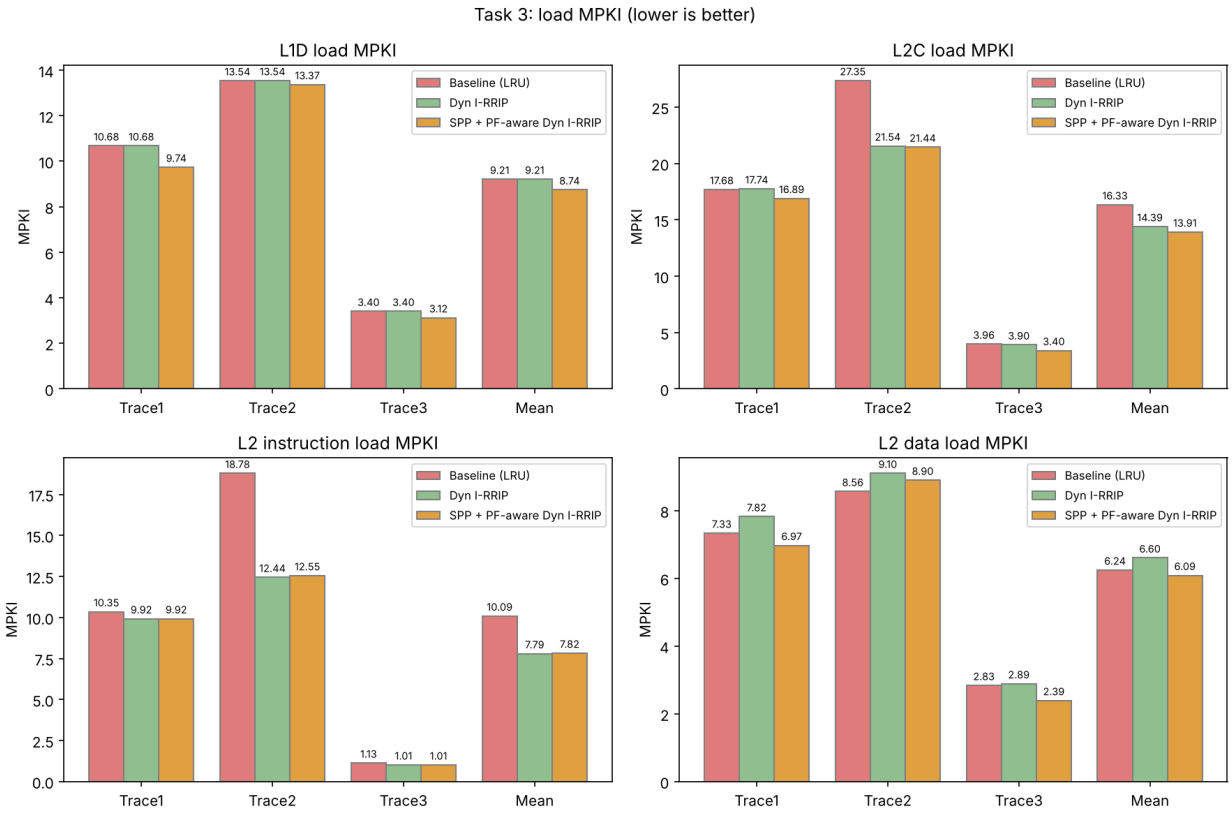


Figure 2: Task 3: L1D, L2C, L2 instruction and L2 data load MPKI

1.3 Observations

- **Task 3 is the best configuration on every trace.** Geomean speedup rises from 1.0078 (Dyn I-RRIP alone) to 1.0136, and every trace improves over Dyn I-RRIP: Trace1 0.9988 → 1.0064, Trace2 1.0250 → 1.0269, Trace3 0.9997 → 1.0075.
- **The instruction protection from Task 2B is preserved.** L2 instruction MPKI stays essentially unchanged versus Dyn I-RRIP (mean 7.79 → 7.82) even though SPP adds tens of thousands of prefetch fills to the L2. Distant insertion, no promotion on prefetch hits, and evicting unused prefetched lines first keep these fills from displacing code.
- **Data misses fall below the baseline.** Mean L2 data MPKI drops from 6.60 (Dyn I-RRIP) to 6.09, lower even than the LRU baseline (6.24). SPP removes data misses that I-RRIP had traded away, so the instruction gain no longer comes at the cost of data.
- **Traces 1 and 3 gain most from prefetching.** They have regular data access patterns: SPP removes 8.8% and 8.3% of L1D load misses, and L2C load MPKI drops by about 5% and 13% versus Dyn I-RRIP.
- **Trace2 gains least from prefetching** (coverage 1.25%). It is dominated by instruction misses (L1I MPKI 40), which a data prefetcher cannot remove; its speedup comes mostly from the replacement policy.
- **Accuracy.** SPP's own filter reports 59–69% accuracy, so throttling keeps the degree mostly at 3 (with some epochs at 2 and 4). ChampSim's metric is lower (27–50%) because it only counts prefetches that missed in the L1D; many SPP prefetches target lines already present.
- **Why prefetch awareness matters at L2.** Only about 3% of prefetched lines filled into the L2 are ever demanded there, which directly supports inserting them at a distant RRPV and evicting them first.